CP115
Computer Programming 1
Semester 1
Session 2026 / 2027
2 hours 30 minutes

Name: _____ **Practicum:** _____

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For Examiner's Use		
Question No.	**Mark**	
	Examiner	**KP/KKP**
1 (24)		
2 (12)		
3 (14)		
4 (14)		
5 (16)		
6 (10)		
TOTAL 90		

This question paper consists of 13 printed pages.

1. In problem solving, a systematic approach helps programmers organize their ideas, identify errors and develop efficient solutions. A team applies this approach to develop a campus bicycle rental program.

   (a) Identify the problem-solving step represented by each activity.

   [*3 marks*]

Activity	Problem-solving step
i. Determine the rental inputs, charging rules and required receipt details.	
ii. Plan the calculation steps using pseudocode before writing Python code.	
iii. Prepare a user guide explaining the inputs and meaning of the receipt.	

   (b) Programming errors can prevent a program from running or cause it to produce incorrect results.

   i. Describe syntax error

   [*1 marks*]

   _____

   _____

   ii. List **two (2)** types of errors other than mentioned in **Question 1.b(i).**

   [*2 marks*]

   _____

   _____

   (c) The Input-Process-Output (IPO) model consists of three components.

   i. Describe the output component.

   [*2 mark*]

   _____

   _____

   ii. For each scenario below, identify whether the activity described represents **input, process or output** in the Input-Process-Output (IPO) model.

   [*6 marks*]

Scenario	Input, Process or Output
A student enters the number of hours spent using a rented bicycle into a campus rental application.	
To arrange parcel delivery, the customer selects a delivery option from the available choices.	
Before confirming a bicycle rental, the student enters the required rental duration into the application.	
During a donation campaign, the program adds each contribution to the running total.	
A learning application determines whether a student qualifies for a bonus voucher based on completed sessions and quiz scores.	
On the application's dashboard, the total number of books donated appears as a numerical summary.	

(d) During a reading campaign, a student enters the number of books read in a month. The program determines the reward points and achievement message according to the table below.

Number of books	Reward Points	Achievement message
0–2	3	"Keep reading"
3–5	8	"Good progress"
6 or more	12	"Outstanding reader"

The program displays the reward points and the achievement message. Assume the number of books is a non-negative integer.

Construct an Input-Process-Output (IPO) table for this program.

[*5 marks*]

Input:	
Process:	
Output:	

(e) After each completed trip, a Gribe driver enters the fare earned in ringgit. The number of trips is unknown. Data entry ends when **−1** is entered. The program calculates and displays the total fares earned and the number of trips with fares exceeding RM20. The sentinel value must be excluded from all calculations. Assume all other inputs are positive.

Construct an Input-Process-Output (IPO) table for this program. Include initialization, the repetition condition and the required updates in the Process column.

[*5 marks*]

Input:	
Process:	
Output:	

2. An algorithm translates a problem's requirements into a logical sequence of precise instructions. Representing it through pseudocode or a flowchart allows programmers to evaluate the solution before implementation.

(a) Briefly describe pseudocode and flowchart. [*2 marks*]

i. Pseudocode:

_____

_____

ii. Flowchart:

_____

_____

(b) A laboratory prepares a rectangular floor mat. The user enters its length and width in metres. The material costs RM18 per square metre. The following algorithm fragments are out of order.

Label	Fragment
A	Print area, cost
B	cost = area * 18
C	Read length, width
D	STOP
E	area = length * width
F	START

Write the labels in the correct execution order. [*3 marks*]

_____

(c) A different program monitors the temperature of a storage room. Study the pseudocode below.

```
START
Read temperature
if temperature > 8 then
    Print "Cooling inspection required"
endif
Print "Reading recorded"
STOP
```

(i) Draw the equivalent flowchart. Include both decision branches and show how execution continues when no inspection message is required. [*5 marks*]

(ii) Complete the output table. Write all messages in the order in which they appear.

[*2 marks*]

Temperature input	Messages printed in order
8	
9	

3. A programming paradigm establishes the principles that guide how programs are structured and solutions are expressed. Different paradigms offer distinct approaches to organizing data, defining operations and controlling program execution.

(a) For each description, identify the programming paradigm. [*2 marks*]

Description	Programming paradigm
i. A solution is organized as an ordered series of procedures that operate on data.	
ii. A solution models bicycles as objects that combine their attributes and behaviours.	

(b) State the most suitable Python data type for each value: a registration code that must preserve leading zeros, such as "0074"; and a flag indicating whether a booking is confirmed.

[*2 marks*]

_____

_____

(c) A program contains the following independent faults. For each fault, name the type of error and write a corrected statement. Treat each statement separately. [*4 marks*]

Fault	Type of error	Corrected statement
i. print("Cost: " + cost) cost = 12.5		
ii. area = length + width		

(d) For a college sports day, an organiser prepares a rectangular banner. Write a complete Python program that reads the banner's length and width in metres and the printing price per square metre. Calculate the banner's area and printing cost. A service charge of 8% of the printing cost and a fixed delivery fee of RM12 are added to determine the total amount payable.

Display the banner's area, printing cost, service charge and total amount payable to two decimal places with appropriate labels. Assume all inputs are positive.

[*6 marks*]

4. (a) Determine the output of each expression. Assume x = 8 and y = 12.

[*6 marks*]

Expression	**Output**
(i) x == y	
(ii) x > y - 5	
(iii) x <= y - 4	
(iv) x != y and x < 10	
(v) x > 10 or y >= 12	
(vi) not (x < 5 or y > 15)	

(b) A theme park determines the entrance fee based on a visitor's age. Visitors aged 12 years and below pay RM15, while visitors older than 12 years pay RM30. **FIGURE 1** shows a Python code segment used to determine the entrance fee.

```
age = int(input("Enter age: "))

if age <= 12:
    fee = 15
else:
    fee = 30

print("Entrance fee: RM", fee)
```

FIGURE 1

Modify the selection control structure in FIGURE 1 by reversing the condition and rearranging the statements without changing the original program requirements.

[*3 marks*]

(c) A smart greenhouse uses Internet of Things (IoT) sensors to monitor soil moisture. The irrigation duration is determined based on **TABLE 1**.

TABLE 1: Irrigation Duration

Soil Moisture Level (%)	**Irrigation Duration (Minutes)**
Below 30	15
30 to 60	10
Above 60	0

Write a complete Python program that accepts the soil moisture level, determines the irrigation duration and displays the result. Assume the input is valid.

[*5 marks*]

5. (a) A counter-controlled repetition structure consists of three main components: initialization, condition and counter update.

Identify the three main components based on the following Python code segment.

```
for counter in range(2, 11, 2):
     print(counter)
```

[*3 marks*]

Part	**Statement**
(i) Initialization	
(ii) Condition	
(iii) Update the counter	

(b) A sports centre records the number of participants attending a training session over five days. FIGURE 2 shows a Python code segment that calculates the total number of participants using a `for` loop.

Convert the following `for` loop into an equivalent `while` loop without changing the original program requirements.

```
total = 0

for counter in range(1, 6):
    participants = int(input("Enter participants: "))
    total = total + participants

print("Total participants:", total)
```

[*5 marks*]

(c) A chef records the cooking time, in minutes, for five dishes prepared in a restaurant. The program calculates the average cooking time.

Complete the following Python code segment by filling in the blanks with appropriate statements.

```
total = 0
counter = 1

while ____________:           # (i)
    cookingTime = int(input("Enter cooking time: "))
    total = total + cookingTime
    counter = _______________    # (ii)

average = ____________        # (iii)
print("Average cooking time:", average)
```

[3 marks]

i. _____

ii. _____

iii. _____

(d) A smart parking system uses an Internet of Things (IoT) sensor to record the number of available parking spaces every hour for 5 hours.

Write a complete Python program using a for loop that accepts the number of available parking spaces for each hour, calculates the total number of available spaces recorded and displays the total. Assume all inputs are valid.

[*5 marks*]

6. A recycling centre records the weight of recyclable materials collected from customers. The program repeatedly asks users to enter the weight of materials in kilograms and the program stops when the user enters -1. Each kilogram of recyclable material earns RM0.80. Customers who submit more than 20 kg in a single transaction receive a bonus of RM5.00 for that transaction.

   Write a complete Python program that:

   - Uses a while loop to accept the weight for each transaction until -1 is entered.
   - Calculates the total weight of all recyclable materials.
   - Counts the number of valid transactions.
   - Counts the number of transactions exceeding 20 kg.
   - Calculates the total payment, including applicable bonuses.
   - Displays the total weight, number of transactions, number of bonus transactions and total payment.

   Assume at least one valid transaction is entered and all valid weights are positive.

   Sample output:

```
Enter weight (-1 to stop): 15
Enter weight (-1 to stop): 25
Enter weight (-1 to stop): 10
Enter weight (-1 to stop): 30
Enter weight (-1 to stop): -1

Total weight: 80.00 kg
Total transactions: 4
Bonus transactions: 2
Total payment: RM74.00
```

[*10 marks*]

End of question paper.